**Ex 1.** (28/04/2026)

Consider a European call option on a forward with pay-off $(F_T - K)_+$ with $F_T = S_T$ forward price at maturity $T$ and $K$ strike

Then the value of this option is given by the Black formula:

$$C(0) = P(0,T)\,(F_0\, N(d_1) - K\, N(d_2))$$

with

$$d_1 = \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{F_0}{K}\right) + \frac{1}{2}\sigma^2 T\right), \quad d_2 = \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{F_0}{K}\right) - \frac{1}{2}\sigma^2 T\right)$$

$P(0,T) = e^{-rT}$ discount factor from $t=0$ to $t=T$

$F_0 = S_0 e^{rT}$ current forward price

$\sigma$ is the volatility of the forward $\rightarrow$ $\underline{dF_t = \sigma F_t\, dW_t}$

$N$ is the normal cdf.

(a) Reformulate this option as an exchange option and use Margrabe formula to derive Black formula.

[Hint: $F_t = e^{r(T-t)} S_t$ $(0 \le t \le T)$ and $F_T = S_T$, $F_0 = e^{rT} S_0$

$P(0,T) = e^{-rT}$, $P(T,T) = 1$, $P(t,T) = e^{-r(T-t)}$]

* we first reformulate this to an exchange option.

Consider 2 assets:

$V_1$: $V_1(t) = e^{-r(T-t)} F_t$

$V_2$: $V_2(t) = e^{-r(T-t)} K$

The pay-off of this exchange option (at maturity) is then given by

$$(V_1(T) - V_2(T))_+ = (F_T - K)_+ \quad ✓$$

So the call option on the forward can be rewritten as an exchange option between $V_1$ and $V_2$.

By Margrabe:

$$C(0) = V_1(0)\, N(d_1) - V_2(0)\, N(d_2)$$

with $d_1 = \frac{1}{\Sigma\sqrt{T}}\left(\ln \frac{V_1(0)}{V_2(0)} + \frac{1}{2}\Sigma^2 T\right)$, $d_2 = \frac{1}{\Sigma\sqrt{T}}\left(\ln\left(\frac{V_1(0)}{V_2(0)}\right) - \frac{1}{2}\Sigma^2 T\right)$

$$\Sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho} = \mathrm{Vol}\left(\frac{V_1}{V_2}\right), \quad \rho = \mathrm{Cor}(V_1, V_2)$$

$\Rightarrow$

We compute the necessary values:

$V_1(0) = e^{-rT} F_0 \quad , \quad V_2(0) = e^{-rT} K$

* Dynamics of $V_1$:

$V_1(t) = e^{-r(T-t)} F_t$ , call $f(t,x) = e^{-r(T-t)} x$ , thus $V_1(t) = f(t, F_t)$

Itô-formula: $dV_1 = \frac{\partial V_1}{\partial t} dt + \frac{\partial V_1}{\partial F_t} dF_t + \frac{1}{2} \frac{\partial^2 V_1}{\partial F_t^2} dF_t dF_t$

$$= r \underbrace{e^{-r(T-t)} F_t}_{V_1} dt + e^{-r(T-t)} \underbrace{dF_t}_{= \sigma F_t dW_t} + 0$$

$dV_1 = r V_1 dt + \sigma V_1 dW_t$

$\Rightarrow \sigma_1 = \sigma$

* Dynamics of $V_2$:

$V_2(t) = e^{-r(T-t)} K$ , call $f(t,x) = e^{-r(T-t)} K$ , thus $V_2(t) = f(t, F_t)$

Itô-formula: $dV_2 = \frac{\partial V_2}{\partial t} dt + \underbrace{\frac{\partial V_2}{\partial F_t}}_{=0} dF_t + \frac{1}{2} \underbrace{\frac{\partial^2 V_2}{\partial F_t^2}}_{=0} dF_t dF_t$

$$= r e^{-r(T-t)} K dt$$
$$= r V_2 dt$$

$\Rightarrow \sigma_2 = 0$

* $\Sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2} = \sigma.$

Margrabe thus yields:

$$C(0) = e^{-rT} F_0 N(d_1) - e^{-rT} K N(d_2)$$
$$= P(0,T) \left( F_0 N(d_1) - K N(d_2) \right)$$

where $d_1 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{V_1(0)}{V_2(0)}\right) + \frac{1}{2}\sigma^2 T \right) = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{F_0}{K}\right) + \frac{1}{2}\sigma^2 T \right)$

$$d_2 = \frac{1}{\sigma\sqrt{T}} \left( \ln\left(\frac{F_0}{K}\right) - \frac{1}{2}\sigma^2 T \right)$$

which yields the Black formula as required.

(b) Derive the Black-formula using a change of numeraire.

use constant $r$ interest rate, $dF_t = \sigma F_t \, dW_t^Q$ with $Q$ risk-neutral prob. measure

By the risk-neutral valuation principle we have

$$C(0) = e^{-rT} E^Q[(F_T - K)_+]$$

$$= e^{-rT} E^Q[(F_T - K) 1_{(S(T)>K)}]$$

$$= e^{-rT} E^Q[F_T 1_{(S(T)>K)}] - K e^{-rT} E^Q[1_{(S(T)>K)}]$$

$$= \underbrace{e^{-rT} E^Q[F_T 1_{(S(T)>K)}]}_{=: C^1(0)} - K e^{-rT} Q(S(T)>K)$$

To compute this first term, we will "change the numeraire"

Define $S_0(t) = e^{rt}$ with $dS_0 = r dt$ risk-free asset, then

$$C^1(0) = \frac{1}{S_0(T)} E^Q[F_T 1_{(S(T)>K)}]$$

$$= E^Q\left[\frac{F_T 1_{(S(T)>K)}}{S_0(T)}\right]$$

(i) $(S_0, Q)$ is numeraire pair:

$S_0 > 0$ so it is a numeraire

and $(S_0, Q)$ numeraire pair follows from definition of a risk-neutral probability measure ✓

(ii) Pick a numeraire $N_1 > 0$ s.t. $E^Q\left[\frac{N_1(T)}{S_0(T)}\right] = 1$:

Choose $N_1(t) = \frac{V(t)}{V(0)}$ with $V(t) = e^{-r(T-t)} F_t$, $V(0) = e^{-rT} F_0$

$$E^Q\left[\frac{N_1(T)}{S_0(T)}\right] = E^Q\left[\frac{V(T)}{V(0) S_0(T)}\right] = \frac{1}{V(0)} E^Q\left[\frac{V(T)}{S_0(T)}\right] = \frac{1}{V(0)} \frac{V(0)}{S_0(0)}$$

$(S_0, Q)$ is a numeraire pair thus $\frac{V}{S_0}$ is a $Q$-martingale

$$= \frac{1}{S_0(0)} = 1. \quad ✓$$

(iii) Apply "Change of numeraire" property:

$(N_1, Q_1)$ is a numeraire pair with $\frac{dQ^1}{dQ} = \frac{N_1(T)}{S_0(T)}$

and

$$C^1(0) = N_1(0) E^{Q^1}\left[\frac{F_T 1_{(S(T)>K)}}{N_1(T)}\right]$$

$$= 1 \cdot E^{Q^1}\left[\frac{F_T 1_{(S(T)>K)}}{V(T)}\right] V(0)$$

$$= V(0) E^{Q^1}[1_{(S(T)>K)}]$$

$$= e^{-rT} F_0 \, Q^1(S(T)>K)$$

Using change of numeraire:

$$E^Q[F_T 1_{(S(T)>K)}]$$

$$\Updownarrow$$

$$E^{Q^1}[1_{(S(T)>K)}]$$

which is great!

$\Rightarrow$

We conclude that

$$C(0) = P(0,T)\left( F_0\, Q^1(F_T > K) - K\, Q(F_T > K)\right)$$

* We will now have a look at the dynamics of $F_T$ to compute the probabilities $Q^1(F_T > K)$ and $Q(F_T > K)$.

* Dynamics of $F_T$ under $Q$:

$dF_t = \sigma F_t\, dW_t$ , use Itô lemma with $f(x) = \ln x$ , then

$$d(\ln F_t) = \frac{1}{F_t}\, dF_t + \frac{1}{2}\left(\frac{-1}{F_t^2}\right) dF_t\, dF_t$$

$$= \sigma\, dW_t - \frac{1}{2}\sigma^2\, dt$$

$$\Rightarrow \ln F_t = \ln F_0 + \int_0^t \sigma\, dW_u - \frac{1}{2}\int_0^t \sigma^2\, du$$

$$\Rightarrow \ln F_t = \ln F_0 + \sigma\, W_t - \frac{1}{2}\sigma^2 t$$

$$\Rightarrow F_t = F_0 \exp\left(-\frac{1}{2}\sigma^2 t + \sigma W_t\right) \quad \text{under } Q.$$

$\rightsquigarrow$ Hence $Q(F_T > K) = Q\left(-\frac{1}{2}\sigma^2 T + \sigma W_T > \ln\left(\frac{K}{F_0}\right)\right)$

$$= Q\left(\frac{W_T}{\sqrt{T}} > \frac{1}{\sigma\sqrt{T}}\left(\ln\left(\frac{K}{F_0}\right) + \frac{1}{2}\sigma^2 T\right)\right)$$

with $\frac{W_T}{\sqrt{T}} \sim N(0,1)$

$$= Q\left(\frac{W_T}{\sqrt{T}} > -d_2\right)$$

$$= Q\left(\frac{W_T}{\sqrt{T}} < d_2\right) = N(d_2)$$

* Dynamics of $F_T$ under $Q^1$:

Radon-Nikodym process $Z(t)$ can be found as follows:

$$Z(t) = E^Q[Z \mid \mathcal{F}_t] = E^Q\left[\frac{dQ^1}{dQ} \,\middle|\, \mathcal{F}_t\right] = E^Q\left[\frac{N_1(T)}{S_0(T)} \,\middle|\, \mathcal{F}_t\right]$$

$$= E^Q\left[\frac{V(T)}{V(0)\, S_0(T)} \,\middle|\, \mathcal{F}_t\right] = \frac{V(t)}{V(0)\, S_0(t)}$$ Since $(S_0, Q)$ is numeraire pair

$$= \frac{e^{-r(T-t)} F_t}{e^{-rT} F_0\, e^{rt}} = \frac{F_t}{F_0}$$ Thus $Z(t) = \frac{F_t}{F_0} = e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}$

Thus $\sigma = -\sigma$ in Girsanov thm, thus $W^{Q_1}(t) = W(t) - \sigma t$ Brownian motion under $Q^1$.

We can now rewrite the dynamics of $F_T$ for $W^{Q_1}$

$$dF_t = \sigma F_t\, dW_t = \sigma F_t\,(dW_t^{Q_1} + \sigma\, dt) = \sigma^2 F_t\, dt + \sigma F_t\, dW_t^{Q_1}$$

$\Rightarrow$

Use Itô formula with $f(x) = \ln x$, then

$$d(\ln F_t) = \frac{1}{F_t} dF_t + \frac{1}{2}\left(\frac{-1}{F_t^2}\right) dF_t \, dF_t$$

$$= \frac{1}{F_t}\left(\sigma^2 F_t \, dt + \sigma F_t \, dW_t^{Q^1}\right) - \frac{1}{2F_t^2} \sigma^2 F_t^2 \, dt$$

$$= \frac{1}{2}\sigma^2 dt + \sigma dW_t^{Q^1}$$

$$\Rightarrow \ln F_t = \ln F_0 + \int_0^t \sigma \, dW_t^{Q^1} + \frac{1}{2}\sigma^2 \int_0^t du$$

$$\Rightarrow F_t = F_0 \exp\left(\frac{1}{2}\sigma^2 t + \sigma W_t^{Q^1}\right) \quad \text{under } Q^1.$$

$\rightarrow$ Hence $Q^1(F_T > K) = Q^1\left(\frac{1}{2}\sigma^2 T + \sigma W_T^{Q^1} > \ln\left(\frac{K}{F_0}\right)\right)$

$$= Q^1\left(\frac{W_T^{Q^1}}{\sqrt{T}} > -d_1\right) \quad \text{with } \frac{W_T^{Q^1}}{\sqrt{T}} \sim N(0,1)$$

$$= Q^1\left(\frac{W_T^{Q^1}}{\sqrt{T}} < d_1\right) = N(d_1)$$

* Putting all the computations together we find:

$C(0) = P(0,T)\left(F_0 N(d_1) - K N(d_2)\right)$ as required ✓